Conveyor Belt Efficiency Improvement

CM3603 - Edge AI

Group 28

Group Members

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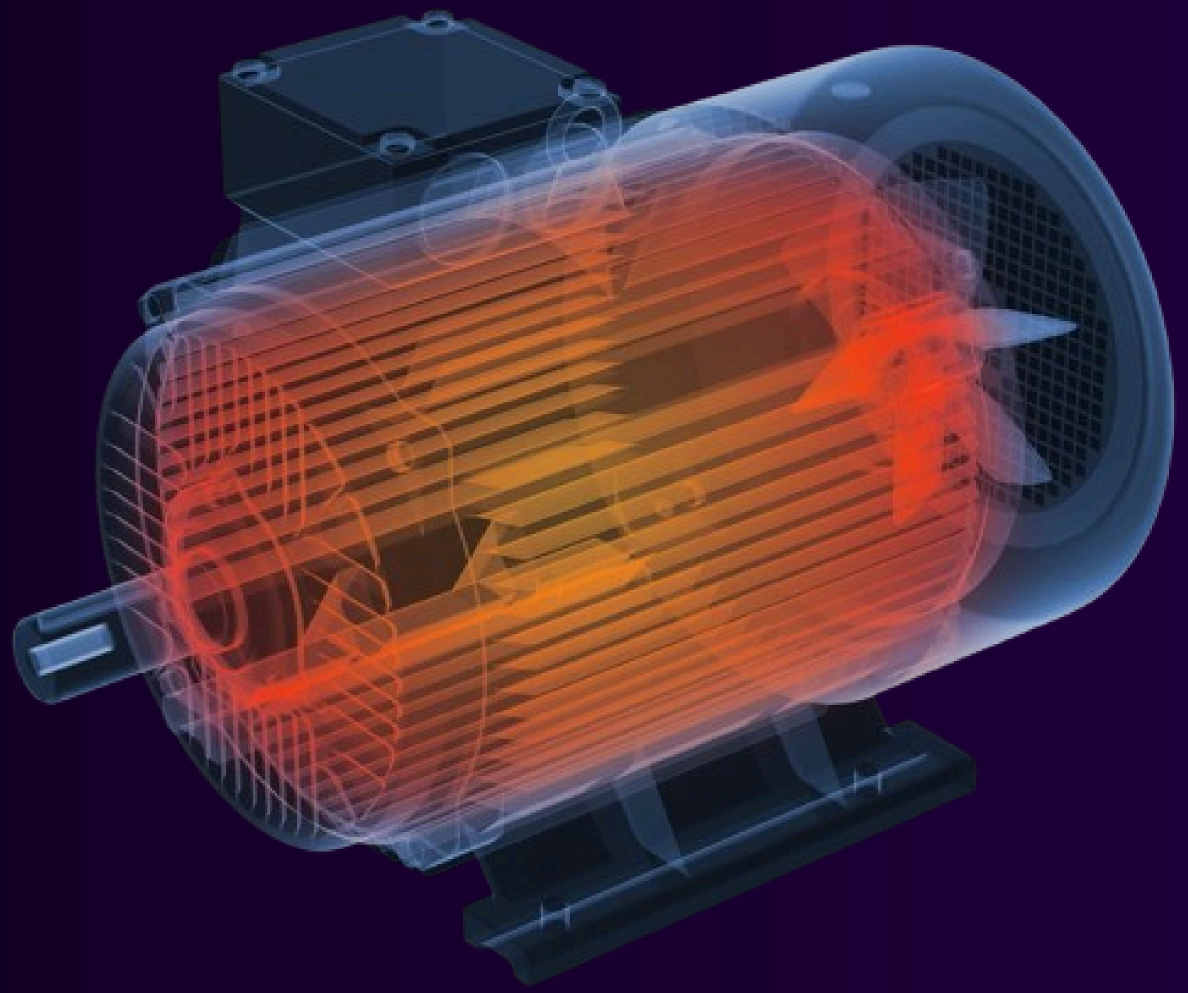
Problem Background

Conveyor belts and assembly lines automate manufacturing with minimal human input, adjusting speeds for production needs and worker safety. Geared motors, essential to these systems, often fail due to overheating or mechanical faults. These failures cause costly downtime, emphasizing the need for smarter, proactive monitoring.

Problem Statement

In mass production industries, the lack of real-time monitoring and predictive analytics for conveyor systems and their geared motors leads to inefficient operational adjustments, undetected motor stress or overheating, and unexpected downtimes that disrupt production flow and increase maintenance costs.

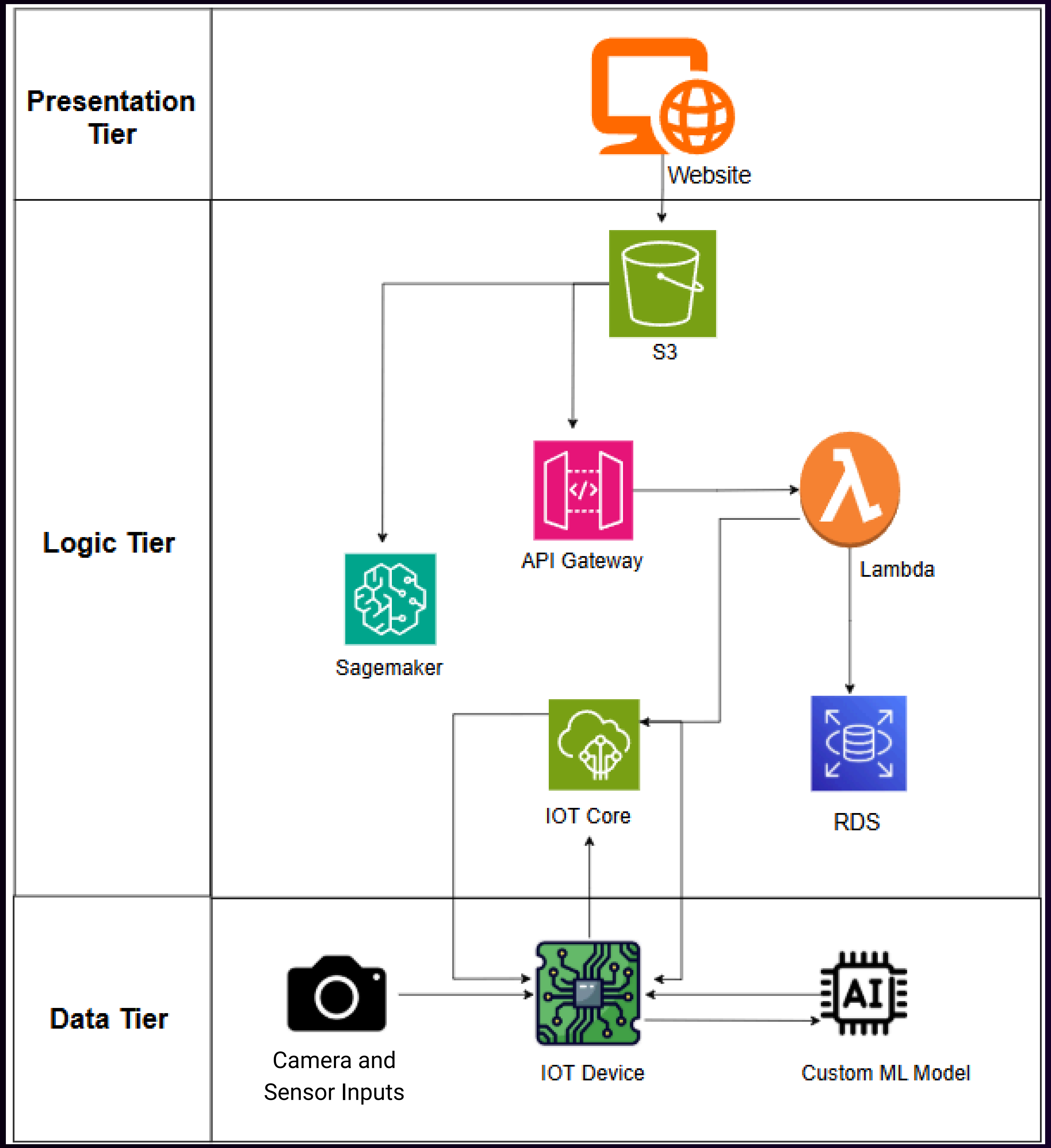


Problem Aim

- This project aims to develop a real-time camera and sensor system to detect, count, track items, and monitor the motor temperature forecast on a conveyor belt.
- Automate the monitoring and data collection process to reduce manual intervention, lower labor costs, and improve throughput in high-speed conveyor belt operations.
- Store and organize captured bottle data in AWS RDS database for seamless integration with inventory management systems, enabling real-time tracking, quality control, and defect detection
- The system will ensure high accuracy, reliability, and scalability to enhance productivity through data-driven decisions.

Demonstration

System Architecture Design



Challenges Faced

1. Preparing a dataset
2. Obtaining Jetson Nano
3. Assembling a conveyor belt
4. Integrating with Cloud Services
5. Python & Ubuntu Dependency issues